

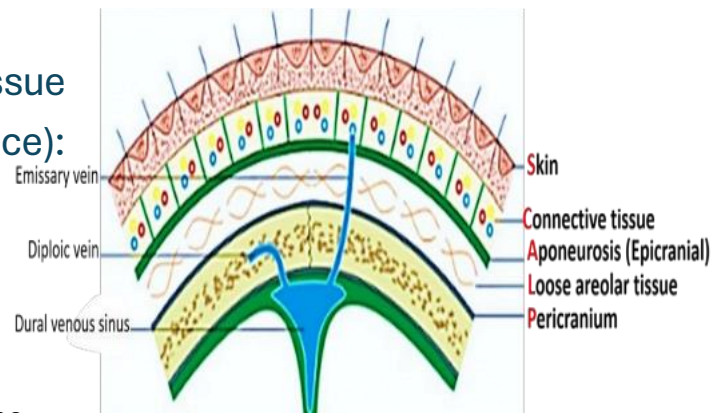
Lecture (1): Anatomy of scalp, face and muscles of the head

Anatomy of the Scalp:

- The scalp includes soft **tissues covering the skull cap**.
- It extends from the **supraorbital margins** anteriorly to the **highest nuchal lines** at the back of the skull and laterally down to the ears and zygomatic arches.
- The **composition** of the scalp is formed of **5 layers**:
 1. **The skin** of the scalp is the thickest and the hairiest in the body. It also contains a high concentration of **sebaceous glands**.
 2. **Dense connective tissue (superficial fascia)**: which firmly unites the **1st & 3rd** layers by **fibrous septa**. The **neurovascular supply** of the scalp is found in this layer.
 3. **Epicranial aponeurosis** is a tough connective tissue sheet whose fibers are oriented in the anteroposterior direction. It connects the two bellies of the occipitofrontalis muscle. Laterally, it blends with the deep temporal fascia & zygomatic arch.

4. Loose areolar tissue (subaponeurotic space):

- It contains **numerous emissary veins** which connect the veins of the scalp to the diploic veins of skull and intracranial venous sinuses.
- It is considered a **dangerous layer** of the scalp as it may be a site of inflammation that may spread from extracranially into intracranially.



5. **Pericranium**: is the periosteum of the vault of the skull.

Occipitofrontalis: consists of occipitalis and frontalis parts.

○ **Occipitalis**:

- Arises from the highest nuchal line of occipital bone.
- Inserted into epicranial aponeurosis. It tenses the aponeurosis.
- It is supplied by the **posterior auricular branch** of facial nerve.

○ **Frontalis**:

- arises from the front of the aponeurosis.

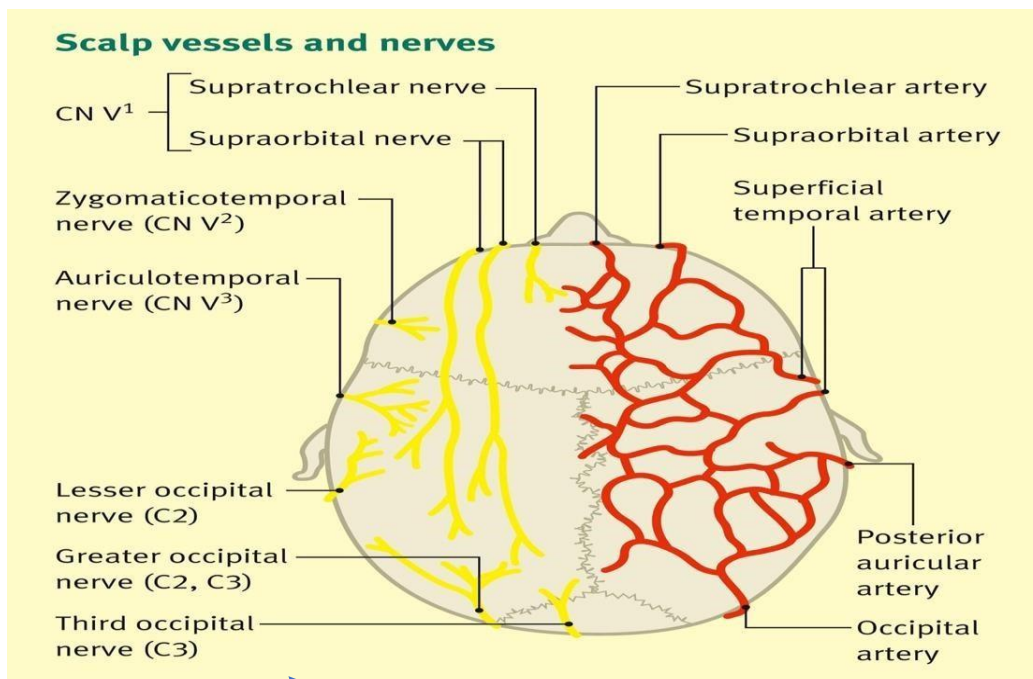
- inserts into upper part of orbicularis oculi and overlying skin of the eyebrow.
- It elevates the eyebrows and produces wrinkles in the skin of the forehead.
- It is innervated by **temporal branch** of facial nerve.

Arterial supply of the scalp: The scalp is supplied by five arteries on each side.

- 3 branches lie in front of the auricle:
 - 1) Supratrochlear artery: a branch of ophthalmic artery.
 - 2) Supraorbital artery: a branch of ophthalmic artery.
 - 3) Superficial temporal artery: from external carotid artery.
- 2 branches behind the auricle:
 - 1) Posterior auricular artery: a branch of external carotid artery.
 - 2) Occipital artery: a branch of external carotid artery.

Sensory nerve supply of scalp:

- the **front of the scalp**: by supratrochlear and supraorbital nerves supply.
- The **temple**: by the auriculotemporal nerve and the zygomaticotemporal nerve.
- Scalp **behind** the ear: by 3rd occipital nerve, greater occipital nerve, lesser occipital nerve and great auricular nerves.



Anatomy of the face

▣ The muscles of the face (muscles of facial expression):

- They are numerous and mainly inserted into the **skin** of the face.
- They are innervated by branches of the **facial nerve**.

1. Orbicularis oculi muscle (sphincter of the eyelids)

- The orbicularis oculi surround the eye socket and extend into the eyelids.
- It is innervated by **temporal** and **zygomatic** branches of the facial nerve.
- It is formed of **three** portions:

1) Orbital portion:

It is surrounding the orbital margins. It arises from the nasal process of frontal bone, frontal process of maxilla and **medial palpebral ligament**. Its fibers form complete ellipse to be inserted into the same point of origin. It closes eyelids tightly.

2) Palpebral portion:

It is confined to the eyelids. It arises from the **medial palpebral ligament** and is inserted into **lateral palpebral ligament**. It closes eyelids gently while blinking and sleeping.

3) Lacrimal portion:

Its fibers arise from the posterior lacrimal crest of lacrimal bone and are attached to fascia surrounding the lacrimal sac. Laterally. They join the upper and lower palpebral fibers. It helps drainage of tears via dilatation of lacrimal sac.

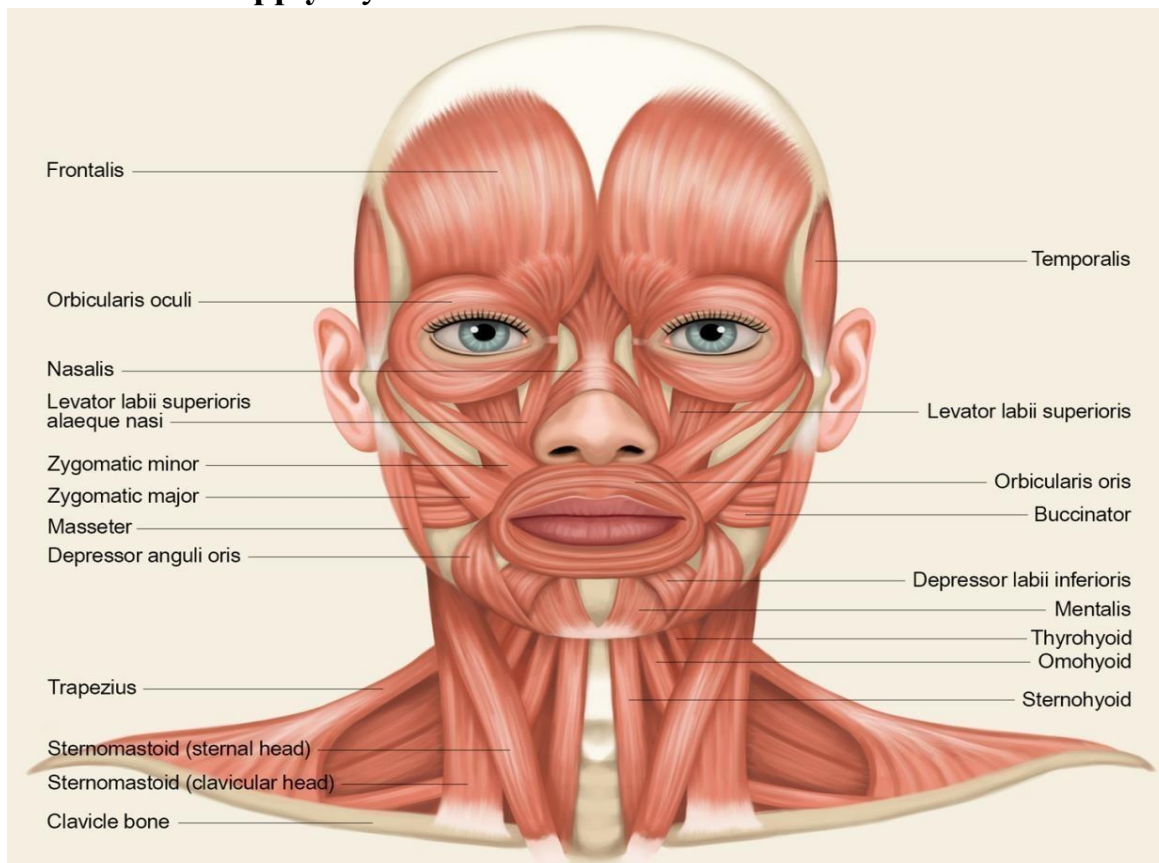
2. Orbicularis oris muscle (sphincter of the mouth):

- The orbicularis oris encircles the mouth, and arises from the maxilla, mandible and surrounding muscles.
- It is inserted into the skin and mucous membranes of lips.
- It contracts, compresses and protrudes the lips.

- It is supplied by **buccal** and **marginal mandibular** branches of facial nerve.

3. Buccinator muscle:

- The **buccinator** forms the muscular aspect of the cheek and lateral wall of the oral cavity.
- **Origin:** arises in part from the pterygomandibular raphe as well as the alveolar processes of both maxilla and mandible.
- **Insertion:** its upper fibers are inserted into the upper lip while the lower fibers into the lower lip, but the central fibers decussate at modiolus lateral to angle of mouth before inserting into upper and lower lips.
- **Action:** It compresses the cheek against the teeth to prevent accumulation of food and saliva in the vestibule of the mouth.
- It is **pierced** by parotid duct & buccal branch of mandibular nerve.
- **Nerve supply:** by buccal branches of the facial nerve.



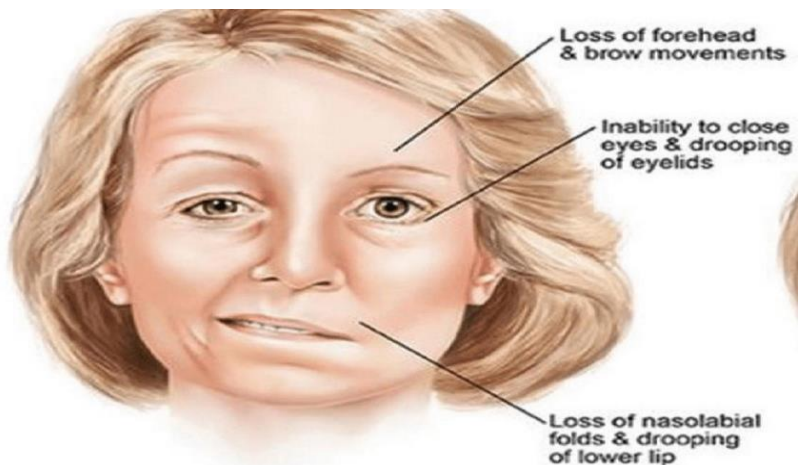
Motor innervation of the face (Facial nerve):

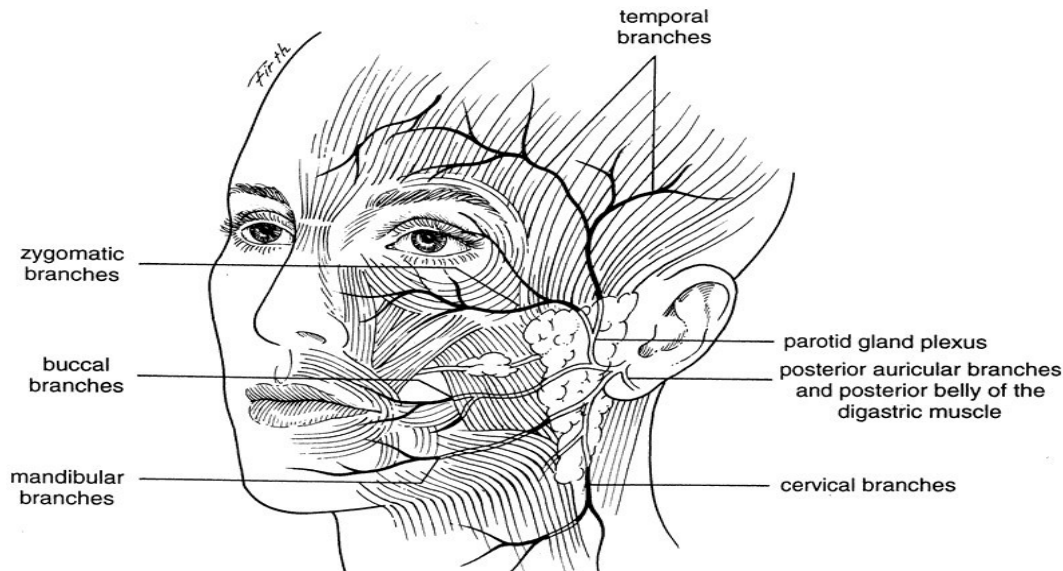
- **facial nerve** emerges from the base of the skull through the **stylomastoid**

foramen.

- just below this foramen, it gives off:
 - 1- **posterior auricular nerve** supplying occipital belly of occipitofrontalis muscle.
 - 2- **Nerve to posterior belly of digastric** supplying this muscle and stylohyoid muscle.
- The **facial nerve** enters the **parotid gland**, lying superficially.
- It gives **five terminal** branches within the gland:
 - 1- The **temporal nerve** (supplying frontalis and orbicularis oculi muscles).
 - 2- The **zygomatic nerve** (supplying orbicularis oculi muscle)
 - 3- The **buccal nerve** (supplying buccinator and muscles of the upper lip).
 - 4- The **marginal mandibular nerve** (supplying muscles of lower lip).
 - 5- The **cervical nerve** (supplying platysma).
- **Lower motor neuron lesion of facial nerve (Bell's palsy):**

- Loss of forehead wrinkles & Inability to elevate eyebrow (paralysis of frontalis muscle).
- Inability to close the eye & drooping of lower eyelid (paralysis of orbicularis oculi).
- Loss of nasolabial fold.
- Deviation of angle of mouth and accumulation of food & saliva in the vestibule of the mouth (paralysis of buccinator & orbicularis oris).





(Facial nerve branches)

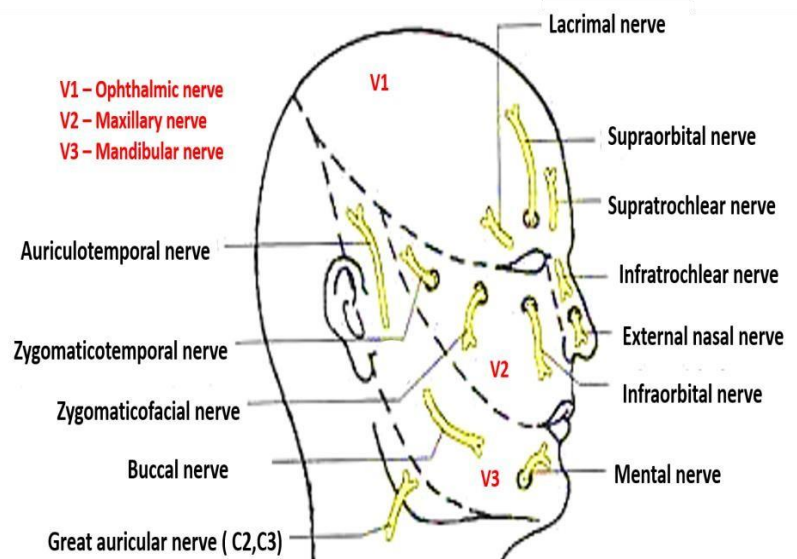
Sensory supply of the face

- The skin of the face is supplied by **(ophthalmic, maxillary and mandibular nerves) trigeminal nerve** except the skin over the angle of the mandible which is innervated by **great auricular nerve from cervical plexus**.

1- Ophthalmic division supplies skin of the forehead, upper eyelid and dorsum of nose through **supra-trochlear, supraorbital, infra-trochlear & external nasal branches**.

2- Maxillary division supplies skin of the lower eyelid, cheek, nose, upper lip and skin overlying zygomatic bone through **infraorbital and zygomaticofacial branches**.

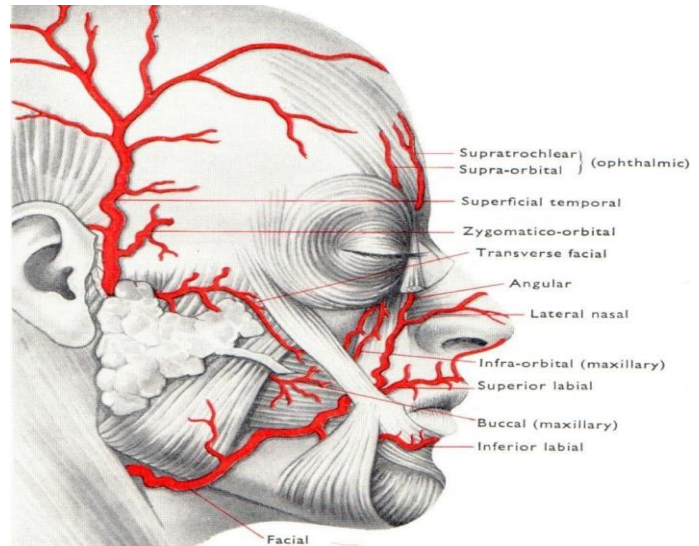
3- Mandibular division supplies skin of the lower lip, chin, skin of the lower jaw through: **auriculo-temporal,**



buccal and mental branches.

■ Blood Supply to the Face:

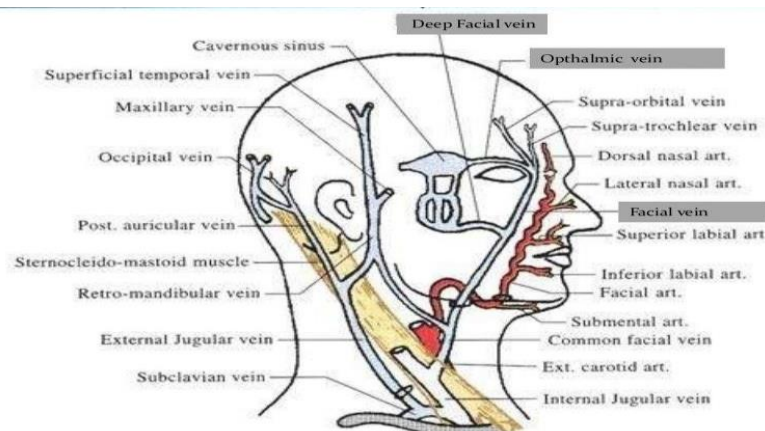
- 1- **The primary arterial supply to the face consists of branches of the facial artery.**
 - The **facial artery** is a branch from the front of **external carotid artery in the neck.**
 - It emerges between the mandible and the submandibular gland.
 - It passes over the inferior margin of the mandible just anteroinferior to the masseter muscle to reach the face.
 - The facial artery has a **tortuous course**. it is always **anterior** to the facial vein. It travels superiorly towards the medial angle of the eye as an **angular artery**.
 - It gives off the following branches in the face: **inferior labial artery, superior labial artery, lateral nasal artery** and its terminal branch (**angular artery**).
- 2- **Transverse facial artery**, from superficial temporal artery.
- 3- **Supraorbital, supratrochlear and dorsal nasal arteries**, from ophthalmic artery.
- 4- **Terminal branches (palpebral, nasal & labial)** of infra-orbital artery.
- 5- **Mental artery** from inferior alveolar artery.



Venous drainage of the face:

- The **supra-orbital** and **supra-trochlear** veins unite to form the **angular vein** which becomes the **anterior facial vein** that has a straight course behind the tortuous facial artery.
- The **anterior facial vein** joins the **anterior division** of the **retromandibular vein** to form the **common facial vein** which drains into the **internal jugular vein**.
- **Superficial temporal vein** joins the **maxillary vein** to form the **retromandibular vein** that passes downwards in the parotid gland.

- It divides into **anterior and posterior divisions**; The posterior division is joined by the posterior auricular vein to form the **external jugular vein**. That descends over sternocleidomastoid muscle and ends into the subclavian vein.
- There is communication between **anterior facial veins** and **ophthalmic veins**.
- Another communication is the **deep facial vein** which connects the anterior facial vein with the pterygoid plexus of veins.
- Spread of infection may occur from the **danger area of the face** between the root of the nose and the angles of the mouth to **cavernous sinus**.



Muscles of mastication:

- The **four** masticatory muscles are **inserted** into the mandible.
- They are supplied by muscular branches of **mandibular nerve**.

Muscle	Origin	Insertion	Action
Masseter	Its superficial head is from the lower border of the zygomatic arch. its deep head from the inner surface of zygomatic arch.	lateral surface of mandibular ramus.	- Elevation and protrusion of the mandible. - Antigravity muscle .
Temporalis	from the temporal fossa. the deep surface of temporal fascia. Fan-shaped muscle.	Coronoid process of the mandible	- Elevation of mandible. - Retraction of the protruded mandible.

Medial pterygoid	Its superficial head from the maxillary tuberosity. Its deep head from the medial surface of the lateral pterygoid plate.	Rough area on the medial surface of the angle of the mandible.	- Elevation and protrusion of the mandible. -Alternating action with lateral pterygoid produces side-to-side movement .
Lateral pterygoid	Its upper head from the infratemporal surface of	• Pterygoid fovea on the front of	• Depression and protrusion of the

	greater wing of sphenoid. Its lower head from the lateral surface of the lateral pterygoid plate.	the neck of the mandible. • the capsule and articular disc of the TMJ.	mandible (mouth open). • Alternating action with medial pterygoid produces side-to-side movement .
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Temporomandibular joint:

➤ It is a synovial joint of condylar variety.

➤ **Articulating bones:**

- **Above:** Mandibular fossa & articular eminence on the undersurface of the temporal bone.
- **Below:** Head (condyle) of the mandible.
- The **articular disc:**

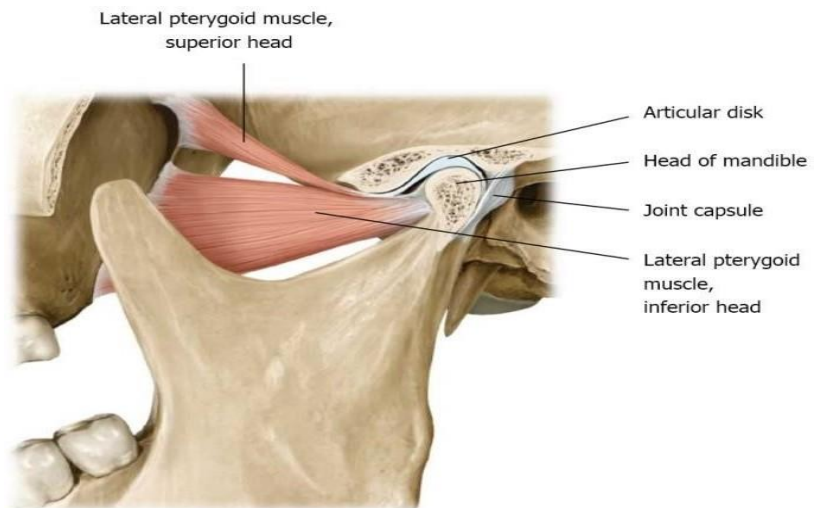
- It is a fibrocartilaginous disc.
- Separates the joint cavity into upper and lower compartments.
- A **gliding movement** occurs in the upper compartment, while a **hinge-like movement** occurs in the lower compartment.

- Its superior surface is anteroposterior concavo-convex.

Inferiorly, the articular disc has a concave surface.

- It is attached around its periphery to the

inside of the capsule and to the **tendon of the lateral pterygoid** anteriorly.



➤ **Ligaments:**

- **Lateral temporomandibular ligament:** strong and thick fibrous band

between the root of zygomatic arch and the neck of mandible.

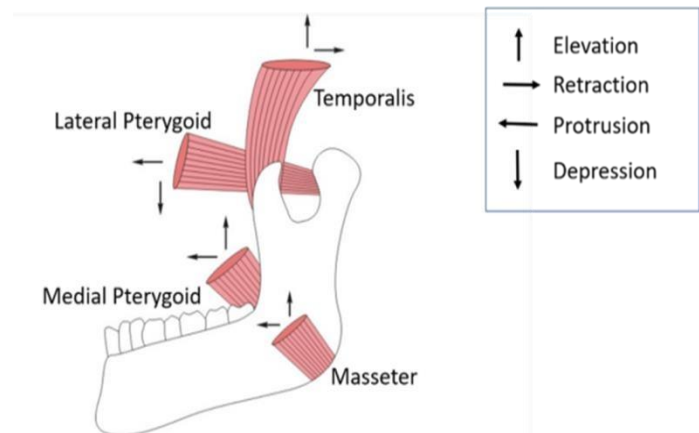
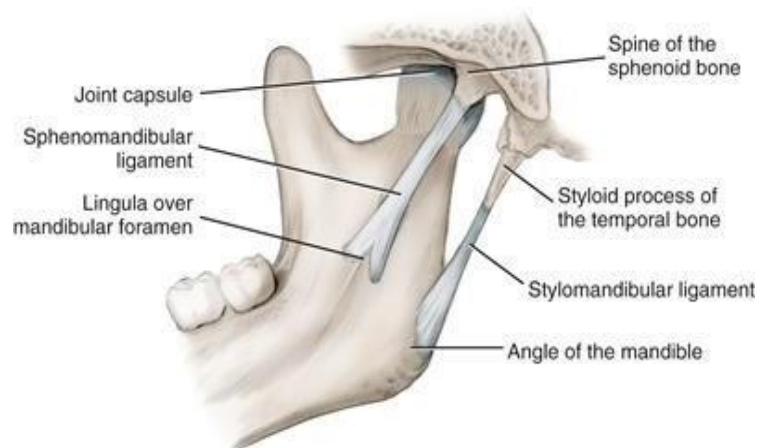
- Pterygo-mandibular ligament.
- Spheno-mandibular ligament.
- Stylo-mandibular ligament.

➤ **Nerve supply:** Auriculo-temporal nerve and nerve to masseter.

➤ **Blood supply:** Superficial temporal artery.

➤ **Movements:**

- **Elevation:** masseter, temporalis & medial pterygoid.
- **Depression:** passively by gravity, actively by the lateral pterygoid, digastric, mylohyoid, geniohyoid and platysma muscles.
- **Protrusion:** by the lateral pterygoid with the aid of the medial pterygoid and superficial head of the masseter.
- **Retraction:** By the horizontal posterior fibers of temporalis muscle only.
- **Side to side movement:** By the medial and lateral pterygoids of both sides acting alternately.



Muscles of the tongue:

- The muscles of the tongue are divided into intrinsic and extrinsic groups.
- The intrinsic muscles are wholly within the tongue and not attached to bone, while the extrinsic muscles have a bony attachment.
- There are four muscles in each group in each half of the tongue, with a midline fibrous septum dividing the organ into two symmetrical halves.
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- The **intrinsic group** are the superior and inferior longitudinal, transverse and

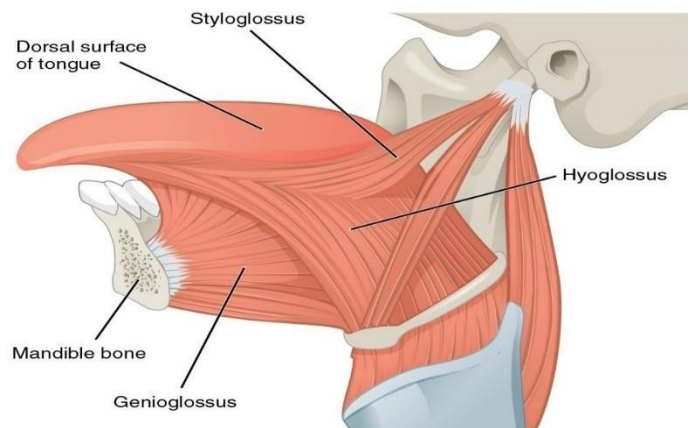
vertical muscles.

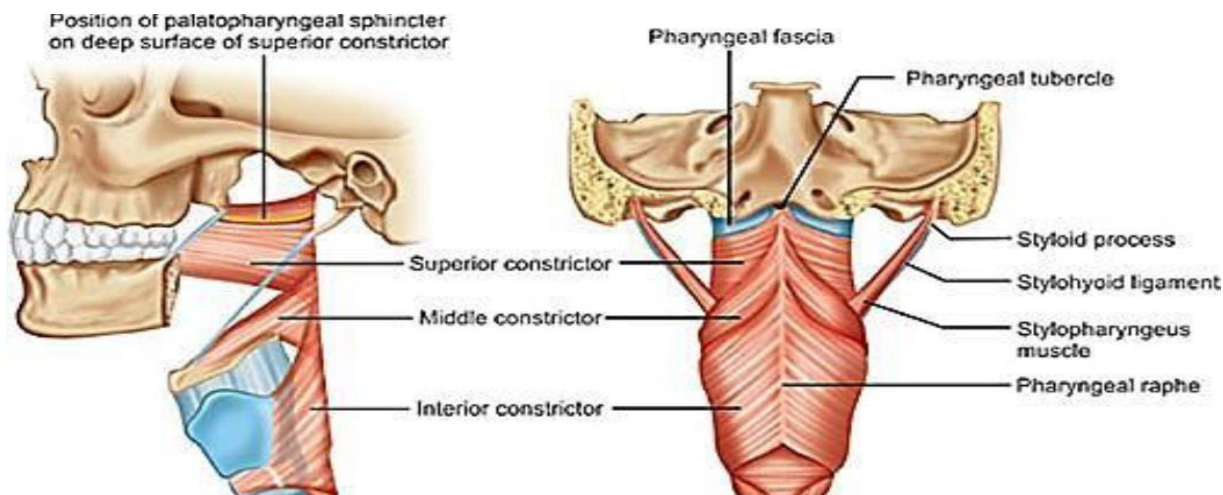
- The **extrinsic group** comprises genioglossus, hyoglossus, styloglossus and palatoglossus.
- All the muscles of the tongue are supplied by the **hypoglossal nerve** except the palatoglossus muscle which is supplied by the **cranial root of accessory through vagus nerve**.

Muscle	Origin	Insertion	Action
Genioglossus	Superior genial tubercle of the mandible.	undersurface of the tongue	Protrudes the tongue.
Hyoglossus	Greater horn of the hyoid bone.	Side of the tongue.	Depresses the tongue
Styloglossus	Styloid process & upper part of stylohyoid ligament.	Side of the tongue	Retracts and elevates the tongue.
Palatoglossus	Undersurface of palatine aponeurosis.	Side of the tongue.	Raises the tongue.

Muscles of the pharynx:

- It consists of:
 - 1- **Three constrictor muscles:** Superior, middle & inferior constrictors which constrict the pharynx during swallowing.
 - 2- **Three longitudinal muscles:** stylopharyngeus, palatopharyngeus & salpingopharyngeus which elevate the pharynx and larynx during swallowing.
- **Superior constrictor muscle** arises from pterygoid hamulus, pterygomandibular raphe and posterior end of mylohyoid line.
- **Middle constrictor muscle** arises from the lower part of the stylohyoid ligament, greater and lesser horns of hyoid bone.
- **Inferior constrictor muscle** arises from oblique line of thyroid cartilage and cricoid cartilage of the larynx.
- The three constrictors have a **common insertion** into the **median pharyngeal raphe** which extends downward from the pharyngeal tubercle Infront of foramen magnum.
- All pharyngeal muscles are supplied by **cranial accessory** through **vagus nerve** except stylopharyngeus which is supplied by **glossopharyngeal nerve**.





Muscles of the soft palate:

- All the **5 muscles** of the soft palate are supplied by the **cranial root of accessory nerve** except **tensor palati** which is supplied by the **mandibular nerve**.

Muscle	Origin	Insertion	Action
Tensor palate	Scaphoid fossa of medial pterygoid plate	Palatine aponeurosis	Tense the palatine aponeurosis and opens auditory tube.
Levator palati	Quadrangle area on the petrous temporal bone	Palatine aponeurosis.	Elevate and pull the palate posteriorly.
Palatoglossus	Inferior surface of palatine aponeurosis.	Side of the tongue.	Raisethe tongue.
Palatopharyngeus	Inferior surface of palatine aponeurosis.	Posterior border of thyroid cartilage.	Elevates the larynx and pharynx, depresses the soft palate.
Musculus uvulae	Posterior nasal spine of the palatine bone.	Mucosa of the uvula.	raises the uvula.

